Bangladesh Fisheries Research Institute
Mymensingh-2201

RESEARCH PROPOSAL

Investigation and Molecular Characterization of Diseases Affecting Cultured Carps in Bangladesh



Fish health · Microbiology · Molecular diagnostics

WHY THIS STUDY

Justification

01

High economic losses and threat to food security

Outbreaks can kill fish overnight, and carps feed millions.

02

Limitations of conventional diagnostic infrastructure

Visible signs and culture plates can't tell look-alike pathogens apart.

03

Mitigating AMR in aquatic ecosystems

Treating without a confirmed diagnosis breeds resistant bacteria.

04

Policy alignment and capacity building

Evidence that supports national AMR, fisheries and SDG goals.



Mass mortality on a pond bank



WHAT WE AIM TO DO

Objectives

Obj. i To investigate the clinical and pathological characteristics of diseases affecting cultured carps

Obj. ii To isolate and identify the major pathogens associated with diseased carps

Obj. iii To determine the molecular diversity and genetic characteristics of the identified pathogens

Obj. iv To analyze the virulence-associated genes of major bacterial pathogens affecting cultured carps



DELIVERABLES

Expected Output

01

Clinical and pathological profiles of major carp diseases

02

Isolation and identification of major carp pathogens

03

Molecular characterization, genetic diversity and virulence gene profiles of major bacterial pathogens

04

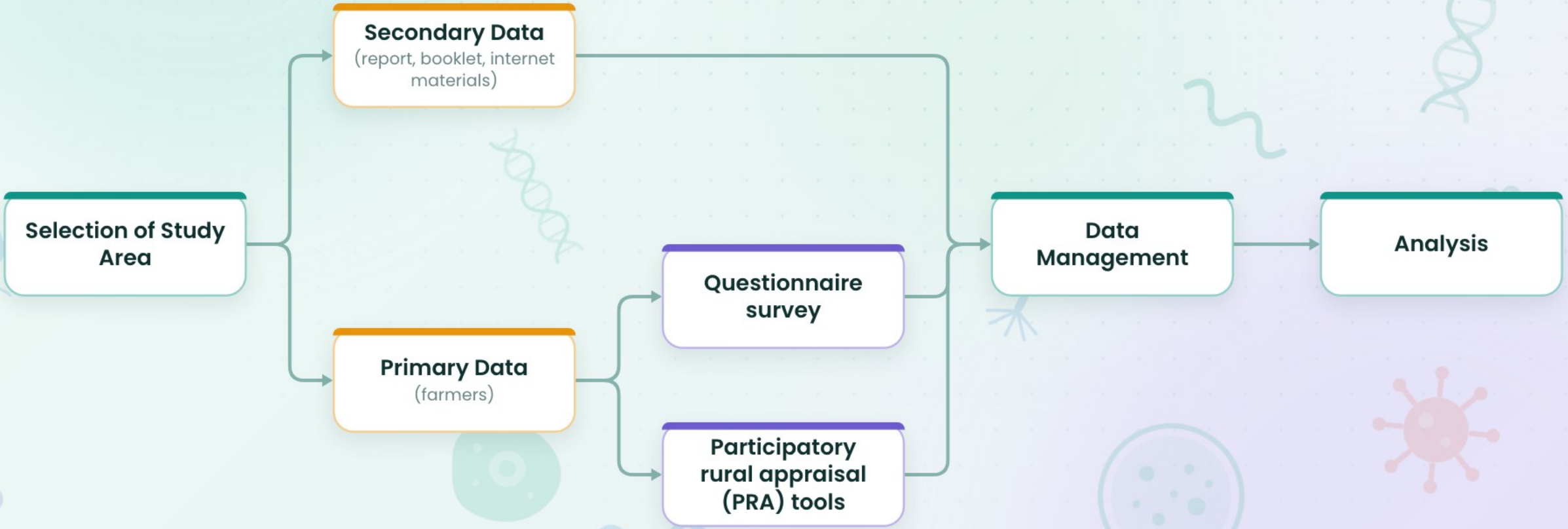
Baseline molecular database of carp pathogens in Bangladesh

05

Evidence-based strategies for disease diagnosis, surveillance and control

ACTIVITY 1

A comprehensive survey to assess farmers' knowledge of diseases of carps



ACTIVITY 1

Survey of Farmers' Knowledge of Carp Diseases

01

Selection of Study Area

Identify freshwater aquaculture regions and carp-farming communities.

02

Data Collection Sources

Primary data: farmers
Secondary data: reports, booklets and internet materials.

03

Questionnaire Survey

Collect farmers' knowledge and experiences of carp diseases.

04

Participatory Rural Appraisal

Use PRA tools to document community knowledge and disease concerns.

05

Data Management

Organize, check and compile the collected survey information.

06

Data Analysis

Analyze survey findings to assess farmers' knowledge of carp diseases.

ACTIVITY 1 · PICTORIAL VIEW OF THE SURVEY METHODOLOGY · FOLLOW STEPS 1–6

Survey of Farmers' Knowledge of Carp Diseases

1



Selection of Study Area

Select freshwater aquaculture regions

2



Primary & Secondary Data

Farmers, reports and published sources

3



Questionnaire Survey

Interview carp farmers

4



Participatory Rural Appraisal

Collect community-based insights

5



Data Management

Organize and validate survey records

6

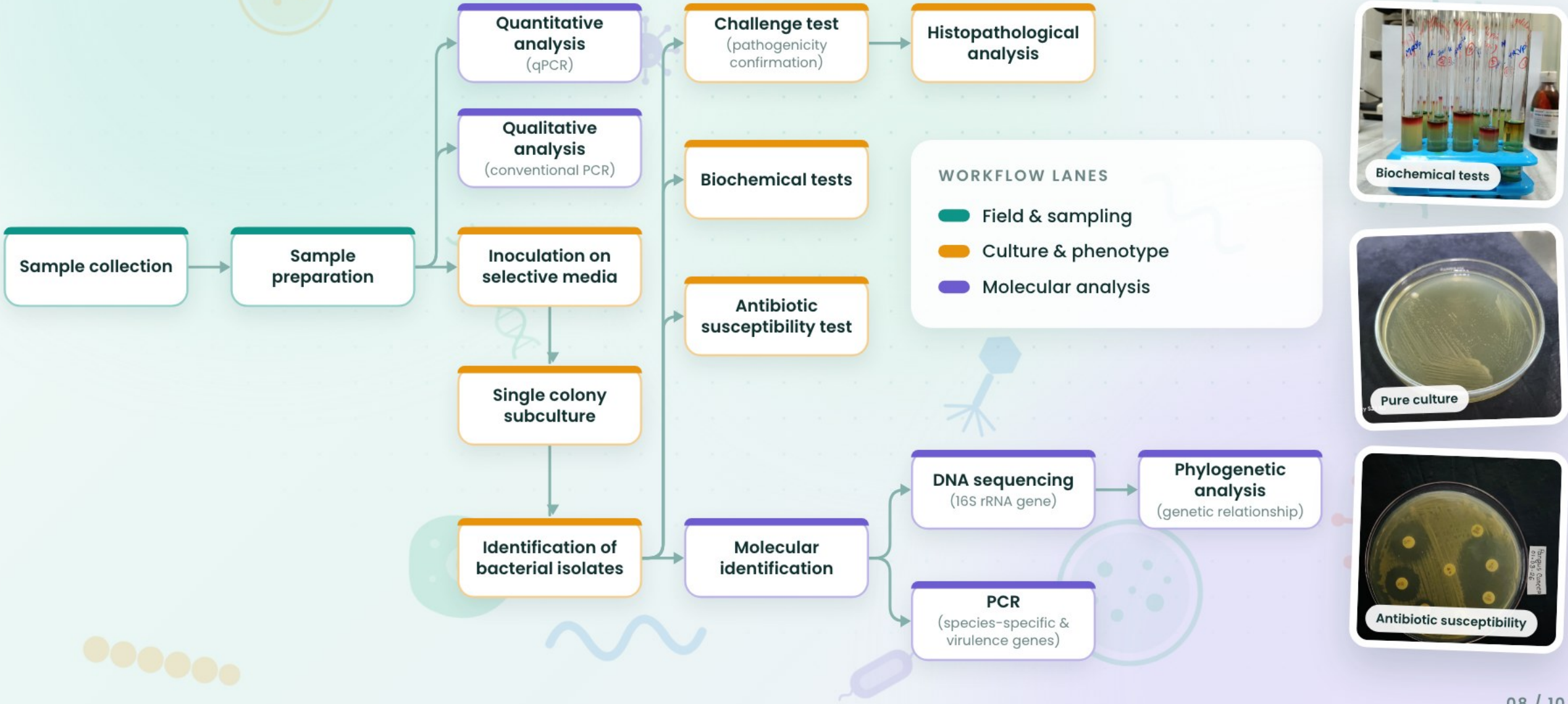


Data Analysis

Interpret findings and patterns

ACTIVITY 2

Isolation and identification of the infectious pathogens in IMCs



TIMELINE · JULY 2026 – JUNE 2027

Work Schedule (2026 - 2027)

YEAR **2026-27**

Project activities	2026 J	A	S	O	N	D	2027 J	F	M	A	M	J
1 A cross-sectional study in selected freshwater aquaculture regions of Bangladesh	Jul – Mar											
2 Sampling of fish from disease affected farms and bacteria isolations	Jul – Mar											
3 Biochemical tests of the isolated pathogens					Nov – Apr							
4 PCR						Dec – Mar						
5 Data analysis & draft writing											May – Jun	

THANK YOU

Questions & discussion are welcome